

# 1. What to fix in the current page

| Current design issue                          | Why it is questionable                                                                                                                                                                 | Recommended design change                                                                                                      |
|-----------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| Treating G/S/R levels as direct impact levels | STPI says the effects column is cautionary and does <b>not</b> claim that specific effects will occur. Actual impacts vary by event and non-space-weather factors.                     | Label scale-based outputs as <b>hazard cues</b> or <b>space weather condition levels</b> , not guaranteed impacts.             |
| Power-grid table relies heavily on Kp/G-scale | STPI says grid users want localized GIC/geoelectric information; GICs directly affect the grid, and E-field in V/km is more useful than Kp alone.                                      | For grid risk, use <b>regional K/Hpo + dB/dt + geoelectric field + local GIC</b> , with Kp as a context indicator only.        |
| “Radio blackout” may imply all radio systems  | STPI says the R-scale reflects x-ray-driven HF radio interference on the sunlit side and does <b>not</b> indicate a total blackout across all frequencies.                             | Rename UI wording to <b>HF Radio Blackout / X-ray HF Absorption</b> and separate it from SATCOM/GNSS scintillation.            |
| Aviation radiation risk tied to S-scale       | STPI says S-scale is based on 10 MeV protons, but aviation dose concern is more tied to much higher energies, around $\geq 500$ MeV; astronauts have different relevant energy ranges. | Split aviation into <b>HF/SATCOM operations</b> and <b>radiation dose</b> , using a D-index/dose-rate product where available. |
| One impact table for many sectors             | STPI says sectors have different needs and some have procedures tied to scale levels, with impacts varying by protections, geography, and infrastructure.                              | Keep the risk matrix, but make the <b>impact definitions sector-specific</b> .                                                 |

Lack of  
uncertainty/confidence

STPI says users often interpret  
scale impacts as certain,  
leading to confusion and  
reaction fatigue; forecasts  
should communicate probability,  
confidence, and uncertainty.

Add confidence/uncertainty text:  
“possible,” “localized,”  
“confidence low/medium/high,”  
and “check sector product.”

## 2. Recommended sector taxonomy for the app

| SWIFT<br>sector/tab            | Keep,<br>split, or<br>merge? | Primary risk question                                                                        | Primary<br>indicators/product<br>s                                                                                                 | Impact focus                                                                                   |
|--------------------------------|------------------------------|----------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| <b>Power Grid</b>              | Keep                         | Could geomagnetic activity produce operationally significant GICs?                           | Regional K/Hpo, dB/dt, geoelectric field V/km, local magnetometers, GIC monitors, Dst for storm context                            | Voltage regulation, reactive power, transformer heating, protection trips, restoration posture |
| <b>Aviation Operations</b>     | Split from generic comms     | Could polar/high-latitude operations require reroute, altitude change, or comms contingency? | G-scale/Hpo for geomagnetic disturbance, S-scale/proton products, HF absorption products, SATCOM coverage, route latitude/altitude | Polar reroutes, fuel planning, HF/SATCOM availability, dispatch/ATC coordination               |
| <b>Aviation Radiation Dose</b> | New sub-tab                  | Could crew/passenger dose exceed operational or regulatory concern?                          | Dose-rate/D-index style product, proton spectrum, ≥500 MeV protons, route, altitude, latitude                                      | Crew/passenger exposure management, altitude reduction, route adjustment                       |

|                                           |                                           |                                                                                                   |                                                                                                                                                    |                                                                                                                          |
|-------------------------------------------|-------------------------------------------|---------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| <b>HF<br/>Communications</b>              | Keep,<br>separate<br>from<br>aviation     | Could HF skywave links<br>degrade or black out?                                                   | R-scale/x-ray flux,<br>D-region absorption,<br>polar cap<br>absorption, solar<br>proton event data                                                 | Loss/degradation of<br>HF on sunlit side or<br>polar routes; backup<br>comms activation                                  |
| <b>Satellites /<br/>Spacecraft</b>        | Keep                                      | Could the environment<br>affect spacecraft health,<br>attitude, charging, drag,<br>or operations? | S-scale/proton flux<br>by energy, electron<br>flux, G/Hpo, neutral<br>density/drag<br>products, radiation<br>belt products                         | Single-event upsets,<br>charging, star tracker<br>issues, solar array<br>degradation,<br>drag/orbit prediction<br>errors |
| <b>GNSS /<br/>Navigation /<br/>Timing</b> | Keep                                      | Could ionospheric<br>conditions degrade<br>accuracy, integrity, or<br>timing?                     | TEC, TEC<br>gradients,<br>scintillation indices,<br>regional ionosphere<br>products, solar radio<br>burst alerts, G/Hpo<br>as context              | Position error, loss of<br>lock, timing<br>degradation, integrity<br>alarms, differential<br>correction<br>degradation   |
| <b>Pipelines</b>                          | Keep, but<br>mark as<br>lower<br>maturity | Could geomagnetically<br>induced currents affect<br>cathodic protection or<br>corrosion risk?     | Geoelectric field,<br>dB/dt, local ground<br>conductivity,<br>pipeline<br>orientation/length,<br>cathodic protection<br>readings                   | CP anomalies,<br>corrosion-risk<br>management,<br>inspection/monitoring<br>posture                                       |
| <b>Human<br/>Spaceflight</b>              | Keep                                      | Could radiation or<br>geomagnetic conditions<br>affect crew safety or<br>mission timeline?        | Proton flux by<br>energy, dose-rate<br>products, $\geq 30$ –50<br>MeV proton<br>products for<br>EVA/thin shielding,<br>mission<br>orbit/trajectory | EVA constraints,<br>storm shelter posture,<br>timeline modification,<br>crew dose<br>management                          |

|                                              |                                             |                                                                                      |                                                                                               |                                                                                    |
|----------------------------------------------|---------------------------------------------|--------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| <b>Emergency Management / Critical Comms</b> | Consider adding                             | Which communications and timing systems remain reliable in the affected region?      | HF absorption, SATCOM/GNSS scintillation, regional products, power-grid status, public alerts | Backup comms, situational awareness, coordination with utilities and public safety |
| <b>Public / Aurora Visibility</b>            | Keep, but do not frame as “risk” by default | Is aurora likely to be visible, and is any public preparedness messaging needed?     | Kp/regional auroral oval, cloud/visibility if integrated, public communication status         | Aurora visibility, public interest, avoiding over-warning                          |
| <b>Space Domain Awareness</b>                | Add or merge with satellites                | Could drag/uncertainty degrade tracking, conjunction assessment, or catalog quality? | Neutral density, G/H <sub>po</sub> , thermosphere/drag products, tracking residuals           | Orbit prediction uncertainty, collision-risk uncertainty, maneuver planning        |
| <b>Surveying / Geomatics</b>                 | Optional specialist tab                     | Could geomagnetic or GNSS disturbance reduce survey quality?                         | K/H <sub>po</sub> , local magnetic field, GNSS scintillation/TEC                              | Magnetic survey postponement, GNSS precision degradation                           |
| <b>Rail</b>                                  | Low-confidence from STPI                    | Possible timing/comms/navigation effects, but STPI had limited sector feedback       | GNSS timing, communications, power-grid dependency                                            | Do not over-specify without rail stakeholder validation                            |
| <b>Agriculture / Precision Agriculture</b>   | Low-confidence from STPI                    | Could GNSS precision degrade automated/precision operations?                         | GNSS integrity, TEC/scintillation, correction-service status                                  | Guidance/positioning degradation                                                   |
| <b>Precision Drilling</b>                    | Low-confidence from STPI                    | Could magnetic/GNSS disturbances affect directional drilling/surveying?              | Local magnetic field, K/H <sub>po</sub> , GNSS integrity                                      | Survey deferral, magnetic reference degradation                                    |

STPI listed many of these end-user communities, but it also says it was less successful in gathering feedback from **rail, agriculture, and precision drilling**, so those rows should be treated as lower-confidence until validated with users.

### 3. Sector impact tables to use in the app

#### Infrastructure and ground systems

| Sector            | 1 — Little /<br>None                                                | 2 — Minor                                                       | 3 —<br>Moderate                                                                                | 4 — High                                                                                             | 5 — Extreme                                                                                                      |
|-------------------|---------------------------------------------------------------------|-----------------------------------------------------------------|------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|
| <b>Power Grid</b> | Routine monitoring; no measurable GIC or voltage deviation expected | Localized high-latitude GIC possible; operator awareness        | Regional voltage-control or reactive-power management may be needed; equipment alarms possible | Significant GIC risk; transformer heating/protection trips possible; postpone vulnerable maintenance | Widespread instability possible; emergency grid operations, restoration coordination, public-safety coordination |
| <b>Pipelines</b>  | No expected CP/GIC effect                                           | Minor cathodic-protection noise possible in susceptible regions | Local CP anomalies; increased monitoring of long conductive assets                             | Widespread CP disturbance possible in exposed networks; corrosion-risk mitigation posture            | Severe induced-current environment; integrity monitoring and post-event review needed                            |

|                                              |                                                                    |                                                              |                                                                                                    |                                                                                               |                                                                                                          |
|----------------------------------------------|--------------------------------------------------------------------|--------------------------------------------------------------|----------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| <b>Emergency Management / Critical Comms</b> | No expected effect on emergency communications                     | Monitor HF/GNSS/SATCOM status                                | Some backup comms or timing services may be degraded regionally                                    | Multiple comms paths may require active management; coordinate with utilities/comms providers | Critical communications or infrastructure dependencies may be unreliable in affected regions             |
| <b>Public / Aurora</b>                       | No unusual public impacts; aurora limited to normal high latitudes | Aurora possible at high latitudes; public interest messaging | Aurora possible farther equatorward; explain space weather without implying infrastructure failure | Widespread public attention; coordinate clear messaging on what is and is not expected        | Exceptional aurora/event messaging; avoid treating public visibility as equivalent to public safety risk |

For the grid and pipeline rows, the key correction is that **E-field/GIC should drive the impact assessment**, with Kp or G/Hpo serving as broad context. STPI specifically notes that GICs directly affect the power grid, that GICs correlate to geoelectric field, and that grid participants asked for an E-field scale/product.

## Aviation, RF, navigation, and space systems

| Sector                     | 1 — Little / None | 2 — Minor                                        | 3 — Moderate                                                                       | 4 — High                                                                         | 5 — Extreme                                                                    |
|----------------------------|-------------------|--------------------------------------------------|------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|--------------------------------------------------------------------------------|
| <b>Aviation Operations</b> | Normal operations | Confer/monitor for polar or high-latitude routes | Consider reroute, SATCOM availability, fuel reserve, or lower-altitude contingency | Plan lower altitude or avoid vulnerable routes; active dispatch/ATC coordination | Active rerouting/descent/route avoidance; sustained comms or radiation concern |

|                                           |                           |                                                       |                                                                                       |                                                                                                |                                                                                                           |
|-------------------------------------------|---------------------------|-------------------------------------------------------|---------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| <b>HF<br/>Communications</b>              | Normal HF conditions      | Minor degradation or short-duration fade possible     | Regional/dayside HF degradation; alternate frequencies or paths may be needed         | Major HF blackout in affected region; backup comms needed                                      | Widespread or prolonged HF loss in affected sector; mission-critical fallback required                    |
| <b>GNSS /<br/>Navigation /<br/>Timing</b> | Normal accuracy/integrity | Minor scintillation or TEC effects; monitor integrity | Accuracy/timing degradation possible; single-frequency and precision users affected   | Wide-area degradation/loss of lock possible; alternate nav/timing needed                       | Severe regional GNSS disruption; mission operations relying on precision/timing may be compromised        |
| <b>Satellites /<br/>Spacecraft</b>        | Routine operations        | Monitor charging/radiation/drag products              | Increased anomaly risk; possible SEUs, star tracker issues, or drag prediction errors | Significant charging, radiation, or drag effects; maneuver/tracking plans may change           | Severe environment for vulnerable assets; loss of function or major orbit-prediction degradation possible |
| <b>Space Domain<br/>Awareness</b>         | Normal tracking quality   | Minor drag/tracking residual changes                  | Increased orbit-prediction uncertainty for LEO                                        | Large conjunction-assessment uncertainty; catalog maintenance burden                           | Severe tracking/orbit-prediction degradation; collision-risk management degraded                          |
| <b>Human<br/>Spaceflight</b>              | Normal mission timeline   | Dose monitoring; no expected crew action              | Elevated review of EVA/timeline/vehicle shielding                                     | EVA restrictions or sheltering posture possible, depending on particle spectrum and trajectory | Crew health-critical radiation posture; shelter/abort EVA/major timeline change                           |

|                                  |                          |                                        |                                                          |                                                                        |                                                                   |
|----------------------------------|--------------------------|----------------------------------------|----------------------------------------------------------|------------------------------------------------------------------------|-------------------------------------------------------------------|
| <b>Surveying /<br/>Geomatics</b> | Normal survey<br>quality | Monitor<br>GNSS/magnetic<br>conditions | Magnetic or GNSS<br>precision survey<br>quality degraded | Postpone<br>sensitive magnetic<br>or high-precision<br>GNSS operations | Severe regional<br>conditions; precision<br>operations unreliable |
|----------------------------------|--------------------------|----------------------------------------|----------------------------------------------------------|------------------------------------------------------------------------|-------------------------------------------------------------------|

The aviation row is based on STPI's airline findings: companies use the SWS for pre- and in-flight polar operations, including conferring with providers at G2/S2 and above, considering reroutes and fuel reserves at stronger levels, planning lower altitude during S4/S5 conditions, and coordinating route changes when radio blackout areas affect planned routing.

The GNSS/SATCOM/radar rows should **not** be driven only by G-scale or R-scale. STPI's phenomenology appendix says ionospheric scintillation can degrade or cause loss of satellite communications, GNSS data, and radar signals, and can produce GNSS positioning/timing errors; it also notes solar radio bursts can affect GNSS, communications, and radar but are not captured by the R-scale.

The spacecraft and SDA rows should include both radiation and neutral-density/drag concepts. STPI says energetic protons can cause charging, single-event upsets, hardware/software issues, false star-sensor readings, intermittent loss of communications, loss of power, and in extreme cases loss of satellite; it also says geomagnetic heating of the upper atmosphere increases drag for satellites below roughly 700 km and increases orbit-prediction uncertainty.

## 4. Risk table: what should drive probability vs. impact

| <b>Hazard/product<br/>family</b>               | <b>Probability should<br/>mean</b>                                                                | <b>Impact should be<br/>selected from</b>                                  | <b>Best-fit sectors</b>                                                          |
|------------------------------------------------|---------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| <b>G-scale / Kp /<br/>Hpo</b>                  | Chance of geomagnetic<br>storm conditions reaching<br>a defined level in a<br>defined time window | Sector-specific<br>consequences; do not<br>assume G-level equals<br>impact | Grid, pipelines,<br>satellites, GNSS,<br>SDA, aurora,<br>emergency<br>management |
| <b>Regional K /<br/>regional H /<br/>dB/dt</b> | Chance of local/regional<br>magnetic-field<br>disturbance exceeding<br>operational triggers       | Grid, pipeline, magnetic<br>survey, regional<br>comms/GNSS impact          | Grid, pipelines,<br>surveying,<br>emergency<br>management                        |



|                                              |                                                                                          |                                                                                               |                                                       |
|----------------------------------------------|------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|-------------------------------------------------------|
| <b>Geoelectric field / GIC</b>               | Chance that local E-field or measured/estimated GIC exceeds sector thresholds            | Grid and pipeline operational consequences                                                    | Power grid, pipelines                                 |
| <b>S-scale / proton flux</b>                 | Chance of energetic proton environment reaching relevant energy/flux thresholds          | Satellite radiation effects, polar HF effects, human dose only if energy spectrum supports it | Satellites, aviation, human spaceflight, polar comms  |
| <b>Dose-rate / D-index product</b>           | Chance route/altitude/trajectory dose exceeds operational threshold                      | Aviation or crewed-spaceflight radiation consequences                                         | Aviation radiation, human spaceflight                 |
| <b>R-scale / x-ray HF absorption</b>         | Chance of dayside HF absorption from solar x-ray flare                                   | HF comms impacts, not all radio systems                                                       | HF comms, aviation, emergency comms, maritime/amateur |
| <b>Scintillation / TEC / T-scale product</b> | Chance of regional ionospheric irregularity affecting trans-ionospheric signals          | GNSS/SATCOM/radar degradation                                                                 | GNSS, SATCOM, radar, emergency management, SDA        |
| <b>Solar radio burst product</b>             | Chance of receiver-facing solar radio noise affecting systems                            | GNSS/comms/radar interference                                                                 | GNSS, SATCOM/comms, radar, aviation                   |
| <b>Neutral density / drag product</b>        | Chance of thermosphere/drag conditions exceeding orbit-maintenance or catalog thresholds | Satellite drag, reentry, conjunction-assessment consequences                                  | Satellites, SDA                                       |

STPI explicitly discusses several new or improved product families that line up with this table: aviation dose/D-index, geoelectric field/E-scale, ionospheric scintillation/T-scale, neutral density, and sector-specific/domain-based alerts.

# 5. Practical recommendation for SWIFT UI

I would keep the overall layout, but change the content model from:

**Sector → NOAA scale threshold → deterministic impact**

to:

**Sector → hazard family → localized indicators → possible impact → suggested action → uncertainty/confidence**

A better sector table schema would be:

| Field                    | Example for Power Grid                                               | Example for Aviation                                                                                |
|--------------------------|----------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| Sector                   | Power Grid                                                           | Aviation Operations                                                                                 |
| Relevant hazard families | Geomagnetic storm, geoelectric field, GIC                            | Geomagnetic storm, proton event, HF absorption, SATCOM/GNSS disruption                              |
| Primary indicators       | Regional K/Hpo, dB/dt, E-field V/km, GIC, Dst                        | G/S/R context, route latitude, altitude, HF absorption, proton energy spectrum, SATCOM availability |
| Impact statement         | “Localized GIC may require voltage/reactive-power monitoring”        | “Polar route may require HF/SATCOM contingency or reroute planning”                                 |
| Action statement         | “Notify grid ops; evaluate maintenance posture; monitor GIC/E-field” | “Confer with provider/ATC; evaluate route, fuel, altitude, comms contingency”                       |
| Confidence               | Based on local measurements and forecast uncertainty                 | Based on forecast confidence, route geometry, and dose/comms products                               |

That is much closer to STPI’s recommendation to replace generic effects language with **risk-informed action statements** for key sectors such as aviation, grid operators, and emergency managers.